



## **Bridging the Digital Divide: Multilingual Youths' Growth Mindset and Emotions in AI-IDLE Engagement**

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#### **Data availability statement**

The raw data cannot be available due to privacy/ethical restrictions.

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**Competing interest statement**

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## **Bridging the Digital Divide: Multilingual Youths' Growth Mindset and Emotions in AI-IDLE Engagement**

### **Abstract**

The burgeoning development of Artificial Intelligence (AI) in language education has yet to fully address motivational and affective complexities shaping digital language learning engagement among multilingual youths in under-representative Global South contexts. Employing a structural equation modelling approach, this study surveys 369 Pakistani tertiary multilingual (Punjabi-Urdu-English) youths to explore the predictive impacts of a growth L2 mindset and L2 emotions on AI-IDLE engagement. The research findings reveal that participants' growth L2 mindset positively and significantly predicts their sense of enjoyment, while their growth L2 mindset lacks predictive power over L2 anxiety and L2 boredom. The results also demonstrate that participants' L2 enjoyment can partially mediate the association between the growth L2 mindset and AI-IDLE engagement and simultaneously make a positive and direct impact on AI-IDLE engagement. Drawing on these quantitative insights, this study offers pedagogical implications, particularly in terms of sustaining the context-specific and multilingual youths to engage with AI-mediated digital learning activities beyond the classroom while mobilising affective dimensions in the crossroad of AI technologies affordances and global English.

### **Keywords**

Informal digital learning of English (IDLE), Artificial Intelligence (AI), growth mindset, L2 enjoyment, L2 boredom, L2 anxiety, Pakistan, multilingual youths

## 1. Introduction

The pervasive digital technologies and transnational interactions have rendered many opportunities for exposure to the target language beyond classrooms (Liu et al., 2025a; 2025b). Situated in the backdrop of English as a lingua franca, informal digital learning of English (IDLE) has recently gained momentum in the Teaching English to Speakers of Other Languages (TESOL) field, with a focus on how L2 English learners assemble their digital devices and online resources to engage in independent and autonomous language learning activities outside the classroom (Lee, 2022; Soyoof et al., 2023). Many factors have been corroborated to have potential roles in IDLE, including motivational factors (Liu et al., 2025a; 2025b, Zou et al., 2025c), affective factors (e.g., enjoyment and boredom) (Zou et al., 2025a), and personality trait-like ones (Lee, 2022). Language learning success depends on both momentary pleasure and unwaveringly prolonged efforts (Khajavy et al., 2021; Liu et al., 2025a; 2025b). Recognising the recent breakthroughs and numerous benefits of Generative Artificial Intelligence (GenAI), GenAI technologies enrich learners' language learning activities outside the classroom by providing adaptive, personalised, and prompt interactions or feedback (e.g., playing an IELTS speaking examiner role and brainstorming ideas or outlines for L2 writing practices) (Wu et al., 2025). Some scholarships have incorporated AI technologies into IDLE, thereby formulating an emerging subfield of IDLE (i.e., AI-mediated informal digital learning of English) (AI-IDLE) (Liu et al., 2025a; 2025b).

While the literature has investigated the predictive roles of affective variables on AI-IDLE (Liu et al., 2025a, 2025b), less attention has been paid to how learners' growth L2 mindset functioned as a motivational factor mobilises L2 emotions to invest in AI-IDLE engagement outside classrooms. While previous studies have demonstrated significant connections between (AI-) IDLE and L2 emotions (Lee & Drajerati, 2019; Liu et al., 2025a; Zou et al., 2025a), research is scarce on the simultaneous effects of positive and negative emotions, and the growth L2 mindset on the AI-IDLE. Furthermore, to our knowledge, motivational and emotional dimensions within AI-IDLE research have received limited scholarly attention, particularly concerning multilingual populations. Notably, as a representative nation of the Global South, Pakistan faces significant challenges amid digital transformation, including infrastructural constraints such as uneven internet accessibility (Shamim, 2008). Addressing these disparities is critical for reducing digital inequities across linguistically diverse communities and bridging the digital divide in Pakistan (Rashid et al., 2018). Given these research gaps, the present study employed a structural equation modelling approach to investigate the predictive impacts of growth L2 mindset and L2 emotions on AI-IDLE among Pakistani multilingual youths. Hence, we propose the following research questions to address these gaps.

**RQ1:** In what ways, if any, does the growth L2 mindset influence Pakistani multilingual youths' L2 emotions?

**RQ2:** In what ways, if any, does their growth L2 mindset negotiate with emotions to invest in AI-IDLE?

## 2. Literature review

### 2.1 AI-mediated informal digital learning of English (AI-IDLE)

Positioned at the intersection of computer-assisted language learning (CALL) and out-of-class language learning (Lee, 2022; Liu et al., 2025b; Zou et al., 2025b), IDLE has garnered increasing scholarly attention over the past decade. This is largely due to its capacity to delineate how L2 learners engage in self-directed, self-guided, and unstructured English learning through multiple digital tools and online platforms (Liu et al., 2025b; Zou et al., 2025a). Building on a growing body of IDLE research (e.g., Lee, 2022; Soyoof et al., 2023, 2025d), AI-IDLE has been conceptualized as “how L2 English learners leverage AI technologies for self-guided and self-directed language learning beyond the classroom” (Liu et al., 2025a, p. 2). Empirical findings from recent studies have substantiated that learners’ engagement in AI-IDLE yields a range of linguistic, motivational, affective, and intercultural benefits, including increased language learning enjoyment (Liu et al., 2025a), enhanced motivation (Liu et al., 2025a). These findings underscore the pivotal role of AI-IDLE in contemporary L2 learning contexts, wherein learners actively employ AI technologies to access, curate, and personalize digital learning resources, and to receive immediate and adaptive feedback (Liu et al., 2025a; 2025b).

In recent years, many issues have been documented, encompassing limited financial resources, poor internet connectivity, and frequent power outages in Pakistan, the context of this study (Shamim, 2008). Language education, in particular, has endured many issues involving unequal access (Rashid et al., 2018), inadequate quality of teacher training (Malik, 2024), and the utilization of conventional pedagogical approaches that lead to diminished student engagement (Shamim, 2008). Given these deficiencies in formal education, IDLE may present an alternative for learners pursuing more accessible, engaging, and self-directed learning opportunities (Rashid et al., 2018). Since limited studies related to IDLE have been undertaken in Pakistan, especially in AI-IDLE environments. There is an absence of research focusing on how Pakistani language learners negotiate their motivational and affective factors to influence their AI-mediated IDLE engagement. Thus, our study attempts to address this research gap.

## ***2.2 Control-value theory***

The Control-value Theory (CVT) constitutes a foundational framework that elucidates the complex interplay between various academic emotions, their antecedents, and their consequences within educational contexts (Pekrun & Perry, 2014). CVT proposes a tripartite taxonomy for classifying academic emotions based on three dimensions: valence, activation, and object focus (Pekrun & Perry, 2014). Within this theoretical framework, L2 enjoyment is characterized as a positive, activating, and activity-focused emotion (Dewaele & MacIntyre, 2016), whereas L2 anxiety is conceptualized as a negative, activating, and outcome-focused emotion (Li & Wei, 2023). In contrast, L2 boredom is identified as a negative, deactivating, and activity-focused emotional experience (Li, 2021).

L2 enjoyment refers to “a complex emotion, capturing interacting dimensions of challenge and perceived ability that reflect the human drive for success in the face of difficult tasks” (Dewaele & MacIntyre, 2016, p. 216). This affective construct has been shown to facilitate various aspects of language learning, including heightened learner engagement (Derakhshan et al., 2024;), enhanced linguistic proficiency (Li et al., 2023), and increased willingness to communicate (Lee, 2022). Empirical evidence consistently supports the positive predictive role of L2 enjoyment in shaping IDLE behaviors (Liu et al., 2025a; 2025b). Given its significant impact on second language acquisition (SLA), it is essential to further investigate the antecedents and consequences of L2 enjoyment outside classrooms, particularly in the

context of AI-mediated environments. In contrast, L2 anxiety, one of the most extensively studied negative emotions in SLA (e.g., Amini Farsani & Seyedshoja, 2025), is typically characterized by communication apprehension, test anxiety, and fear of negative evaluation (Horwitz, 2017). Although prior research has established a negative correlation between L2 anxiety and academic achievement (Botes et al., 2020), its influence on learners' engagement in IDLE contexts remains insufficiently explored. The third academic emotion, L2 boredom, has received comparatively limited scholarly attention within the field of SLA (Li et al., 2024). It is defined as a negative and deactivated emotion (Li et al., 2021). Empirical research has demonstrated that L2 learners who frequently experience boredom during the language learning process tend to exhibit diminished concentration, which in turn contributes to lower levels of academic engagement and a tendency to disengage from learning tasks (Li et al., 2021). Accordingly, examining how L2 enjoyment, anxiety, and boredom, influence learners' engagement in AI-IDLE represents a critical issue to be investigated.

### ***2.3 Empirical findings between growth L2 mindset, emotions, and AI-IDLE***

#### ***2.3.1 Research on growth L2 mindset and emotions***

Mindsets are identified as a way of fostering individual learning performance within different realms (Dweck, 1999). Building on the social-cognitive theory proposed by Dweck (1999), mindsets encompass fixed and growth mindset. While fixed mindset refers to a fixed trait that learners believe their language achievement cannot be improved, growth mindset is conceptualized as a changeable trait that learners believe their language competent can be promoted by their effort and perseverance (Khajavy et al., 2021). In other words, learners with a higher level of growth mindset are more likely to invest in time and effort to master a language and to focus on endeavors that they undertake, instead of being afraid of engaging in learning tasks (Wang et al., 2021). In the L2 settings, growth L2 mindset has been evidenced to enhance L2 learners' engagement (Derakhshan et al., 2024), facilitate their willingness to communicate (Wang et al., 2021), promote their language performance (Khajavy et al., 2021), and foster their positive emotions (Wang et al., 2021). Specifically, focusing on the Iranian EFL context, Derakhshan et al. (2024) disclosed that the growth language mindset made a negative, significant, and direct impact on L2 boredom and the growth language mindset influenced student engagement indirectly via the mediating effect of boredom (N = 287). Similarly, another research conducted by Wang et al. (2021) demonstrated that the growth language mindset had an indirect effect on L2 WTC fully mediated by enjoyment, pride, and boredom. While the growth language mindset played a negative and direct role in L2 boredom, it made a positive and direct impact on L2 enjoyment (Wang et al., 2021). More recently, Amini Farsani et al. (2025) found that the growth language mindset had a significant but weak correlation with L2 anxiety (N = 500). Given the close association between growth L2 mindset and academic emotions (i.e., L2 enjoyment, anxiety, and boredom) in prior studies, we pose the following hypotheses.

- H1.*** Growth L2 Mindset negatively influences L2 Boredom.
- H2.*** Growth L2 Mindset negatively influences L2 Anxiety.
- H3.*** Growth L2 Mindset positively influences L2 Enjoyment.

#### ***2.3.2 Research on the growth L2 mindset and AI-IDLE***

While previous research did not validate the predictive role of the growth L2 mindset in AI-IDLE, few studies have confirmed how the growth L2 mindset fosters student engagement

(Fan et al., 2024; Sadoughi & Hejazi, 2023) and the interaction between the growth L2 mindset and learning behaviors (Lou & Noels, 2020). Specifically, employing a multi-stage cluster sampling technique and structural equation modeling approach, Sadoughi and Hejazi (2023) surveyed 295 Iranian EFL learners to explore the relationships between teacher support, growth language mindset, L2 grit, and academic engagement. Their study revealed that the growth language mindset made a positive and direct impact on academic engagement. Likewise, Fan et al. (2024) found that Chinese high schoolers with a high level of growth language mindset are more inclined to engage in language learning (N = 868). Moreover, some variables (i.e., L2 grit and burnout) play mediating roles in the link between growth language mindset and engagement. In the same vein, Zhong et al. (2024) disclosed that language mindset made a positive and significant impact on engagement (N = 1,738). Although previous studies did not examine whether the growth L2 mindset predicts student engagement with AI-IDLE, it can be proposed that the growth L2 mindset is associated with learning behaviors or activities. Given the positive and significant relationship between the growth L2 mindset and student engagement or language achievement (Khajavy et al., 2021; Sadoughi & Hejazi, 2023) and how the growth L2 mindset can function as a motivator of intercultural interactions within traditional classrooms (Lou & Noels, 2020), we tentatively postulate that:

**H4.** Growth L2 Mindset positively influences AI-IDLE engagement.

#### *2.3.3 Research on L2 emotions and AI-IDLE engagement*

L2 emotions include positive emotions (e.g., L2 enjoyment) and negative emotions (e.g., L2 anxiety and boredom), which exert significant roles in language learning process. Reviewing prior studies, we have found studies that investigated the interaction between L2 enjoyment and AI-IDLE (Liu et al. 2025a) and a few studies concentrating on the associations between L2 emotions and IDLE or engagement/performance (Dai & Wang, 2024; Lee, 2022; Zou et al., 2025a, 2025b). Insights from these studies can help formulate hypotheses about the affective dynamics of AI-IDLE. For instance, Liu et al. (2025a) found that EFL students with clearer ideal L2 self and greater FLE, are more likely to engage in AI-mediated language learning activities beyond the classroom (N = 299). In the same vein, focusing on 673 university EFL students from China, Dai and Wang (2024) found that academic emotions play positive roles in student engagement. Similarly, Dewaele et al. (2023) disclosed that L2 anxiety is a negative predictor of academic achievement (N = 332). Although their research did not explore whether L2 learners' negative emotions (i.e., L2 anxiety and L2 boredom) predict students' participation in AI-IDLE activities, it can be concluded that L2 anxiety and boredom are negatively linked with AI-IDLE engagement. Given the established association between L2 emotions and engagement or IDLE in both traditional classrooms and outside the classrooms, this causal pattern may extend to AI-mediated language learning environments. As such, we pose the following hypotheses:

**H5:** L2 Anxiety negatively influences AI-IDLE engagement.

**H6:** L2 Enjoyment positively influences AI-IDLE engagement.

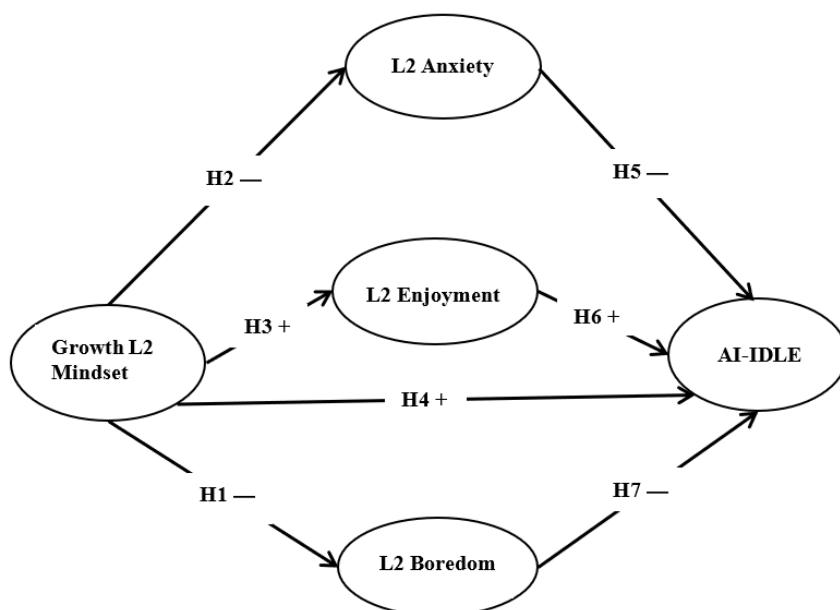
**H7:** L2 Boredom negatively influences AI-IDLE engagement.

Taken together, the seven hypotheses that may be helpful for addressing the two research questions are displayed in Figure 1.

**Figure 1**



*The hypothesized structural model.*



### 3. Methodology

#### 3.1 Research context and participants

Pakistan is a multilingual and multicultural country, with 68 languages spoken within its borders (Ethnologue, 2025). According to the Pakistan Demographic Survey (2001), Punjabi holds the dominant position, with 44.15% of the population speaking it as their primary indigenous language. Other significant regional languages include Pashto, Sindhi, Siraiki, and Balochi. The less prominent indigenous languages of Pakistan are sometimes associated with lower prestige and cultural shame. Urdu serves as Pakistan’s national language, while English has attained the status of an official language (National Education Policy, 2017). English and Urdu occupy the top two positions in the language hierarchy, while the remaining major and minor indigenous languages are relegated to the bottom. English is considered the most influential language due to its strong association with the “domain of power,” encompassing areas such as administration and education (Rahman, 1996). As the English language has attained the status of a lingua franca, L2 English educators and researchers are re-evaluating the psychological factors that shape the terrain of English language education.

In the context of digital technology and education, Pakistan faces significant educational challenges. With a population of 247 million, the demand placed on the English language education system is immense. Numerous issues have been documented in recent years, including the absence of clear policies, limited educational resources, and a range of socioeconomic obstacles (Shamim, 2008). English language instruction struggles with disparities in access (Tayyab et al., 2023), inadequate teacher training and ineffective language pedagogy (Malik, 2024), as well as a reliance on outdated teaching methods that cause low levels of student motivation and engagement (Shamim, 2008). The government has sought to harness technology to address these educational shortcomings (Asad et al., 2021). However, many tertiary institutions lack sufficient infrastructure, with especially rural institutes affected by unreliable internet access and frequent power outages (Taimur-ul-

Hassan, 2013). In addition, many instructors are ill-equipped for or resistant to incorporating technology into their teaching practice (Asad et al., 2021; Taimur-ul-Hassan, 2013). Given these pitfalls in formal education in Pakistan, AI-IDLE may offer an alternative for students seeking more flexible, engaging, and learner-centered pathways, especially since the majority of students (in higher education) already carry mobile phones (Rashid et al., 2018). Thus, we aim to explore how Pakistani students leverage GenAI tools to learn English beyond the classroom.

Conducted within Pakistan's multilingual context, for this study, we recruited participants from seven tertiary institutions using a convenience sampling approach (Creswell & Creswell, 2017). We distributed an e-poster featuring an online questionnaire link and a QR code via WhatsApp, LinkedIn, and Facebook. Collaboration with eight tertiary-level instructors further facilitated survey dissemination to students between November 2024 and March 2025. The participants were selected based on one criterion – namely, current enrolment as a student at a Pakistani tertiary institution.

A total of 369 multilingual learners from tertiary institutions across Pakistan completed the online questionnaire. The sample comprised 42.5% male, 56.9% female, and 0.5% who did not disclose gender. Most participants were aged 18–22 (54.2%), followed by those above 22 (44.4%). Respondents came from diverse academic backgrounds, primarily Social Sciences (35.8%), Science and Engineering (23.6%), and Arts and Humanities (9.2%). The majority studied at public universities (69.7%), with others enrolled in private universities or tertiary institutes. Most participants (67.2%) reported stable Wi-Fi access, and 81.8% had sufficient mobile data for learning. A large proportion of participants (77.5%) reported using ChatGPT for English learning, followed by 16.5% who used other AI tools, while smaller percentages used Claude AI (3.3%) and Bing Chat (2.7%). These demographic and infrastructural profiles offer an essential context for understanding informal language learning among Pakistani multilingual youths.

### ***3.2 Data collection and analysis***

Data were collected via an English-language survey on Google Forms (see Appendix). Participation required informed consent. The questionnaire included two parts: demographic information and 36 items measuring five constructs - growth L2 mindset, L2 enjoyment, L2 boredom, L2 anxiety, and AI-IDLE - using a six-point Likert scale (1 = strongly disagree, 6 = strongly agree).

- (a) Growth L2 mindset (4 items): To evaluate the participants' growth L2 mindset, the current study deployed the growth L2 mindset scale constructed and validated by Papi et al. (2019). The reliability of the growth L2 mindset scale yielded a Cronbach's alpha coefficient of .86, exemplifying excellent internal consistency. An example item was, "No matter how much intelligence you have for learning languages, you can always change it a lot."
- (b) L2 enjoyment (7 items): Building on Liu et al.'s (2024b) 7-item Foreign Language Enjoyment (FLE) scale, we contextualized the instrument to align with informal digital language learning environments. Given that the original scale was designed to measure enjoyment in non-English target language (LOTE) contexts, we adapted the wording of items to reflect the EFL context. For instance, the statement "I am not tired of learning the target LOTE" was revised to "I am not tired of learning English" to ensure

contextual relevance. The refined scale demonstrated robust psychometric reliability, with a Cronbach's alpha coefficient of .90, indicating high internal consistency.

- (c) L2 boredom (11 items): To assess participants' L2 English, the current study employed the Foreign Language Boredom Scale constructed and validated by Li et al. (2024). A Cronbach's alpha coefficient of .94 indicates excellent internal consistency. An example item was: "Studying (English) is dull in general."
- (d) L2 anxiety (6 items): To evaluate the L2 anxiety of the participants, in this study, we deployed the L2 anxiety scale constructed and validated by Papi et al. (2010). A Cronbach's alpha coefficient of .89 exemplifies excellent internal consistency. An example item was, "How afraid are you of sounding stupid in English because of the mistakes you make?"
- (e) AI-IDLE (8 items): The 8-item AI-IDLE scale was constructed and validated by Liu et al. (2024a; 2025a) to evaluate the frequency and diversity of L2 learners' use of generative AI (e.g., ChatGPT, Claude AI, Bing Chat, and others) to help their English learning practices beyond the classroom. The reliability of the overall AI-IDLE scale was .91 in this Pakistani multilingual context, indicating a high index of internal consistency. An example item was, "I use AI technologies to simulate real-life English language use situations beyond the classroom."

Following Collier's (2020) five-step SEM approach, we used SPSS 29.0 and AMOS 29.0 for data analysis. After cleaning the data (e.g., removing incomplete or rapid responses), we assessed normality via skewness and kurtosis, then calculated item means and standard deviations. Reliability was examined using Cronbach's alpha, and validity was tested through confirmatory factor analysis (CFA). Finally, we constructed and validated a structural model to explore the relationships among growth L2 mindset, L2 boredom, L2 anxiety, L2 enjoyment, and AI-IDLE engagement.

#### 4. Research findings

This study reports findings from a comprehensive analysis of data on Pakistani tertiary-level multilingual learners. Descriptive statistics outlined learners' profiles across key variables, followed by reliability and validity checks. Structural equation modeling and mediation analysis were then used to test the hypothesized relationships and address the research questions.

##### 4.1 Descriptive statistics

As presented in Table 1, the descriptive statistics summarize participants' self-reported perceptions regarding their growth L2 mindset, L2 anxiety, L2 enjoyment, L2 boredom, and the frequency of engagement with AI-IDLE activities. The mean scores for items measuring growth L2 mindset ranged from 4.38 to 4.66. For L2 anxiety, item means varied between 3.43 and 3.54. The mean scores for L2 enjoyment items ranged from 4.00 to 4.63, while those for L2 boredom ranged from 3.09 to 3.68. Regarding the frequency of engagement with AI-IDLE activities, item means ranged from 3.89 to 4.21. The kurtosis values of the items ranged from -1.30 to 0.37, and skewness values varied between -0.98 and 0.22. These values fall within the acceptable range of  $\pm 2$ , indicating a satisfactory level of normality in the data distribution (Kunnan, 1998).

**Table 1**  
*Descriptive statistics*

| Factors               | Items | Mean | SD   | Kurtosis | Skewness | Factor loading<br>(In CFA) | $\alpha$<br>( $> .7$ ) |
|-----------------------|-------|------|------|----------|----------|----------------------------|------------------------|
| Growth_L2_M<br>indset | GM1   | 4.66 | 1.30 | 0.05     | -0.82    | 0.76                       | .86                    |
|                       | GM2   | 4.53 | 1.30 | -0.01    | -0.74    | 0.86                       |                        |
|                       | GM3   | 4.50 | 1.29 | -0.26    | -0.63    | 0.84                       |                        |
|                       | GM4   | 4.38 | 1.35 | -0.38    | -0.62    | 0.68                       |                        |
| L2_Anxiety            | A1    | 3.45 | 1.56 | -1.03    | -0.06    | 0.59                       | .89                    |
|                       | A2    | 3.43 | 1.52 | -1.02    | 0.01     | 0.69                       |                        |
|                       | A3    | 3.54 | 1.52 | -0.94    | -0.12    | 0.82                       |                        |
|                       | A4    | 3.52 | 1.54 | -1.00    | -0.04    | 0.84                       |                        |
|                       | A5    | 3.52 | 1.55 | -1.04    | -0.07    | 0.81                       |                        |
|                       | A6    | 3.47 | 1.69 | -1.22    | -0.07    | 0.77                       |                        |
| L2_Enjoyment          | F1    | 4.38 | 1.42 | -0.08    | -0.76    | 0.68                       | .90                    |
|                       | F2    | 4.00 | 1.50 | -0.77    | -0.42    | 0.70                       |                        |
|                       | F3    | 4.40 | 1.34 | -0.26    | -0.64    | 0.79                       |                        |
|                       | F4    | 4.63 | 1.33 | 0.37     | -0.98    | 0.81                       |                        |
|                       | F5    | 4.45 | 1.34 | 0.19     | -0.87    | 0.76                       |                        |
|                       | F6    | 4.36 | 1.41 | -0.32    | -0.70    | 0.72                       |                        |
| L2_Boredom            | F7    | 4.52 | 1.39 | 0.26     | -0.90    | 0.78                       | .94                    |
|                       | B1    | 3.14 | 1.62 | -1.11    | 0.22     | 0.79                       |                        |
|                       | B2    | 3.40 | 1.60 | -1.07    | 0.02     | 0.81                       |                        |
|                       | B3    | 3.50 | 1.55 | -0.98    | -0.06    | 0.79                       |                        |
|                       | B4    | 3.31 | 1.61 | -1.11    | 0.02     | 0.81                       |                        |
|                       | B5    | 3.53 | 1.53 | -0.89    | -0.06    | 0.83                       |                        |
|                       | B6    | 3.51 | 1.53 | -0.98    | -0.10    | 0.79                       |                        |
|                       | B7    | 3.60 | 1.52 | -0.99    | -0.11    | 0.72                       |                        |
|                       | B8    | 3.68 | 1.52 | -0.92    | -0.22    | 0.74                       |                        |
|                       | B9    | 3.09 | 1.66 | -1.24    | 0.18     | 0.75                       |                        |
| AI_IDLE               | B10   | 3.24 | 1.61 | -1.14    | 0.06     | 0.80                       | .91                    |
|                       | B11   | 3.43 | 1.73 | -1.30    | -0.001   | 0.67                       |                        |
|                       | AI1   | 4.16 | 1.32 | -0.27    | -0.43    | 0.63                       |                        |
|                       | AI2   | 3.89 | 1.37 | -0.60    | -0.25    | 0.75                       |                        |
|                       | AI3   | 3.91 | 1.42 | -0.70    | -0.32    | 0.73                       |                        |
|                       | AI4   | 4.00 | 1.26 | -0.38    | -0.30    | 0.83                       |                        |
|                       | AI5   | 4.12 | 1.30 | -0.50    | -0.41    | 0.81                       |                        |
|                       | AI6   | 4.06 | 1.29 | -0.50    | -0.32    | 0.81                       |                        |
|                       | AI7   | 4.21 | 1.30 | -0.14    | -0.59    | 0.75                       |                        |
|                       | AI8   | 4.17 | 1.28 | -0.32    | -0.44    | 0.72                       |                        |

#### 4.2 Reliability and validity checking

As it was suggested by Kline (2023), the Cronbach's  $\alpha$  values of five variables were .86 (growth L2 mindset), .89 (L2 anxiety), .90 (L2 enjoyment), .94 (L2 boredom), and .91 (AI-IDLE), which all exceeded the benchmark of .70 (see Table 1), thus indicating a high index of internal consistency.

To evaluate the validity of the constructs, we compute the convergent and discriminant validity. By following Collier (2020), we considered composite reliability (CR) and average variance extracted (AVE) to examine convergent validity. As offered in Table 2, the CRs of the five factors were above the threshold value of .7, and the AVEs are above the cut-off value of .5 (Hair et al., 2009). To evaluate the discriminant validity, we compared the square root of AVE for each construct with their respective inter-factor correlation coefficients (Hair et al., 2009). In Table 2, the square roots of AVE exceeded their corresponding correlation coefficients, suggesting that discriminant validity was achieved.

**Table 2**

*Convergent and discriminant validity of growth L2 mindset, L2 anxiety, L2 enjoyment, L2 boredom, and AI-IDLE.*

|          |                       | AVE<br>(> .5) | CR<br>(> .7) | The Square root of AVE and Squared<br>correlation coefficient matrix |            |            |            |            |
|----------|-----------------------|---------------|--------------|----------------------------------------------------------------------|------------|------------|------------|------------|
|          |                       |               |              | <b>1</b>                                                             | <b>2</b>   | <b>3</b>   | <b>4</b>   | <b>5</b>   |
| <b>1</b> | Growth_L2_<br>Mindset | .62           | .87          | <b>.79</b>                                                           |            |            |            |            |
| <b>2</b> | L2_Anxiety            | .58           | .89          | .11                                                                  | <b>.76</b> |            |            |            |
| <b>3</b> | L2_Enjoyment          | .56           | .90          | .57                                                                  | .14        | <b>.75</b> |            |            |
| <b>4</b> | L2_Boredom            | .60           | .94          | .01                                                                  | .52        | .01        | <b>.77</b> |            |
| <b>5</b> | AI_IDLE               | .57           | .91          | .44                                                                  | .23        | .55        | .14        | <b>.76</b> |

**Note:** The bolded values along the diagonal represent the square root of AVE.

As for the measurement model, we considered five constructs and their respective observed variables for improving the construct validity of the adapted questionnaire in AMOS. To evaluate the degree of model fit between the measurement model and the dataset, we considered the following seven indices: the ratio of  $\chi^2$  to Degrees of Freedom ( $\chi^2/\text{df}$ ), Comparative Fit Index (CFI), Incremental Fit Index (IFI), Tucker-Lewis index (TLI), Parsimony Goodness-of-Fit Index (PGFI), Root Mean Square Error of Approximation (RMSEA). As presented in Table 3, all model indices were obtained based on the recommended ranges, reflecting a good model fit between the measurement model and the dataset. Therefore, the questionnaire data presented a high index of construct validity.

**Table 3**

*Model fit indices of growth L2 mindset, L2 anxiety, L2 enjoyment, L2 boredom, and AI-IDLE.*

|                                | $\chi^2/\text{df}$ | CFI  | IFI  | TLI  | PGFI | RMSEA |
|--------------------------------|--------------------|------|------|------|------|-------|
| The measurement model          | 2.23               | .92  | .92  | .908 | .72  | .06   |
| The structural model           | 2.234              | .916 | .916 | .909 | .735 | .058  |
| Benchmark values (Kline, 2023) | < 5                | > .9 | > .9 | > .9 | > .5 | < .10 |

#### 4.3 The structural equation model and mediation model measurement

Employing AMOS 29, we conducted an SEM analysis to test the proposed hypotheses. As presented in Table 3, the model demonstrated good fit indices, meeting all thresholds recommended by Kline (2023). According to Figure 2 and Table 4, the structural model supported three of the seven hypotheses formulated in this study (i.e., H3, H5, and H6). In particular, the results indicated that the growth L2 mindset significantly and positively predicted L2 enjoyment ( $\beta = .63, p < .001$ ), supporting H3. However, the growth L2 mindset

did not significantly predict L2 boredom ( $\beta = .01, p = .834$ ), L2 anxiety ( $\beta = .10, p = .084$ ), or AI-IDLE ( $\beta = .13, p = .054$ ), leading to the rejection of H1, H2, and H4. Regarding emotional antecedents of AI-IDLE, L2 enjoyment significantly and positively predicted AI-IDLE ( $\beta = .51, p < .001$ ), supporting H6, while L2 anxiety also exhibited a significant but weaker positive relationship with AI-IDLE ( $\beta = .12, p = .029$ ), supporting H5. By contrast, the relationship between L2 boredom and AI-IDLE was not statistically significant ( $\beta = .10, p = .052$ ), rendering H7 unsupported. These findings underscore the influential role of positive emotional experiences, particularly enjoyment, in mediating the effect of a growth L2 mindset on AI-IDLE.

We further deployed a bootstrapping test with 2,000 samples at a 95% confidence level in AMOS to examine the mediating roles of L2 enjoyment, anxiety, and boredom in the relationship between the growth L2 mindset and AI-IDLE. The results indicated a significant indirect effect through L2 enjoyment (95% *CI* [.19, .40],  $p = .001$ ), confirming partial mediation. However, the indirect effects through L2 anxiety (95% *CI* [−.004, .04],  $p = .193$ ) and L2 boredom (95% *CI* [−.01, .02],  $p = .966$ ) were not significant. These results underscore the significant mediating role of L2 enjoyment in the relationship between a growth L2 mindset and AI-IDLE, whereas the mediating effects of L2 anxiety and L2 boredom were statistically non-significant and, therefore, did not contribute meaningfully to the model.

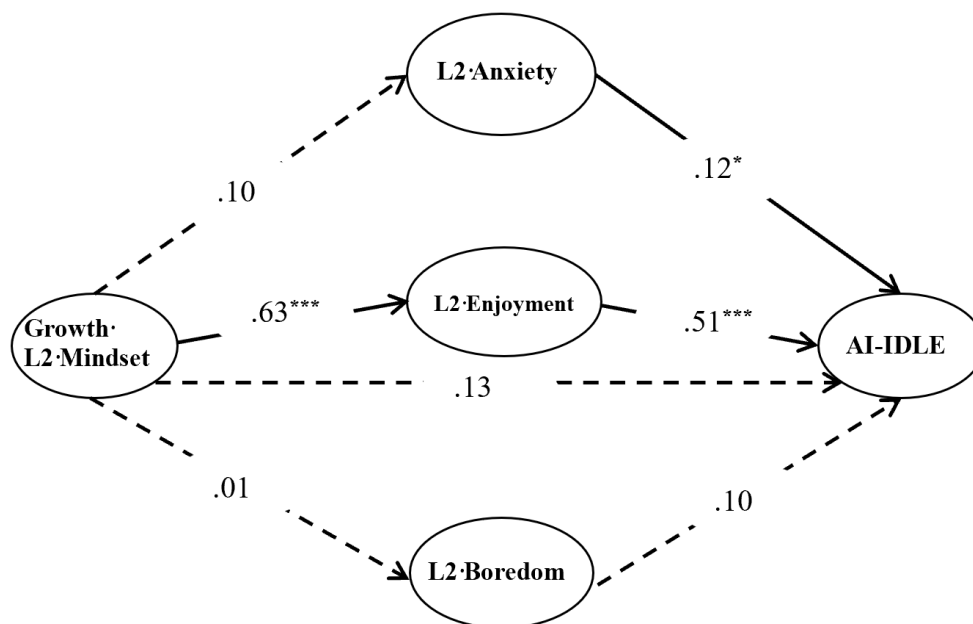
**Table 4**  
*Hypotheses testing results of paths.*

| Hypotheses                                                         | $\beta$                 | $p$         | t-value<br>( $>1.96$ )        | $r$             | Results           |
|--------------------------------------------------------------------|-------------------------|-------------|-------------------------------|-----------------|-------------------|
| H1: Growth L2 Mindset $\rightarrow$ L2 Boredom                     | .01                     | $p = .834$  | .21                           | -.01            | Unsupported       |
| H2: Growth L2 Mindset $\rightarrow$ L2 Anxiety                     | .10                     | $p = .084$  | 1.73                          | .10             | Unsupported       |
| H3: Growth L2 Mindset $\rightarrow$ L2 Enjoyment                   | .63 <sup>***</sup>      | $p < .001$  | 9.57                          | .63             | Supported         |
| H4: Growth L2 Mindset $\rightarrow$ AI-IDLE                        | .13                     | $p = .054$  | 1.93                          | .46             | Unsupported       |
| H5: L2 Anxiety $\rightarrow$ AI-IDLE                               | .12 <sup>*</sup>        | $p = .029$  | 2.18                          | .25             | Supported         |
| H6: L2 Enjoyment $\rightarrow$ AI-IDLE                             | .51 <sup>***</sup>      | $p < .001$  | 6.61                          | .60             | Supported         |
| H7: L2 Boredom $\rightarrow$ AI-IDLE                               | .10                     | $p = .052$  | 1.94                          | .16             | Unsupported       |
| Mediation paths                                                    | 95% Confidence Interval |             | $p$ (Two-Tailed Significance) | Indirect effect | Results           |
|                                                                    | Lower bound             | Upper bound |                               |                 |                   |
| Growth L2 Mindset $\rightarrow$ L2 Anxiety $\rightarrow$ AI-IDLE   | -.004                   | .04         | $p = .193$                    | .011            | No mediation      |
| Growth L2 Mindset $\rightarrow$ L2 Enjoyment $\rightarrow$ AI-IDLE | .19                     | .40         | $p = .001$                    | .28             | Partial mediation |
| Growth L2 Mindset $\rightarrow$ L2 Boredom $\rightarrow$ AI-IDLE   | -.01                    | .02         | $p = .966$                    | .001            | No mediation      |

Note:  $p < .001^{***}$ ,  $p < .01^{**}$ ,  $p < .05^*$ ;  $r < .25$  = small effect size,  $.25 < r < .5$  = medium effect size, while  $r > .5$  = large effect size (Plonsky & Ghanbar, 2018).

**Figure 2**

*The structural equation model of growth L2 mindset, L2 anxiety, L2 enjoyment, L2 boredom, and AI-IDLE.*



Note.  $*p < .05$ ,  $***p < .001$ . The dashed lines indicated non-significant paths ( $p > .05$ ).

## 5. Discussion and pedagogical implications

### 5.1 RQ1: In what ways, if any, does the growth L2 mindset influence Pakistani multilingual youths' L2 emotions?

This study demonstrated that the growth L2 mindset can significantly and positively predict L2 enjoyment, which is aligned with previous studies on how a clear growth mindset pattern can facilitate students to enjoy more language learning journey (Liu et al., 2025a; Zhong et al., 2024). This observation can be reflected in the strong predictive power of the growth L2 mindset on L2 enjoyment. The large effect size ( $r = .63$ ; Plonsky & Oswald, 2014) also confirms the statement that L2 learners who have a higher level of growth L2 mindset are more likely to enjoy language learning beyond the classroom. In addition, the study disclosed that the growth L2 mindset did not predict L2 boredom, diverging from the study of Wang et al. (2021). There are some possible explanations for the finding that the growth L2 mindset lacks predictive power over L2 boredom. One plausible explanation for this novel finding may reside in the intricate nature of boredom as a negative emotion, given that boredom is a multidimensional construct that can be shaped by a myriad of factors beyond mindset, such as learning environment and emotion regulation (Li et al., 2021). Consequently, the absence of a significant association between growth mindset and boredom may suggest that other variables exert a more substantial influence on this emotion in the context of AI-mediated unstructured language learning. An alternative justification could be the lack of pertinence. To be precise, learners possessing the growth L2 mindset may nonetheless encounter boredom if the learning content or activities fail to correspond with their interests, objectives,

or real-world applications. Should learners be unable to discern the worth or relevance of the subject matter, they may experience a sense of disconnection and boredom, irrespective of their conviction in the capacity for growth (Fan et al., 2024).

Moreover, the study indicated that the growth L2 mindset did not make a significant impact on L2 anxiety, leading to the rejection of H2, which diverges from the case of Amini Farsani et al. (2025). This finding suggests that Pakistani multilingual youths may have had limited exposure to a diverse array of learning strategies that align with the growth L2 mindset, thereby constraining its influence on alleviating anxiety. An additional possible reason is that Pakistani multilingual learners may have not been provided with sufficient guidance in effectively tackling learning anxiety even though they have a clear growth L2 mindset image (Lou & Noels, 2020).

### ***5.2 RQ2: In what ways, if any, does their growth L2 mindset negotiate with emotions to invest in AI-IDLE?***

This study has found that while learners' growth L2 mindset did not make a direct impact on AI-IDLE practices, the relationship between their growth L2 mindset and AI-IDLE can be mediated by L2 enjoyment. This finding is not only consistent with the observation of Li (2023) regarding the mediating role of L2 enjoyment in the link between growth mindset and student engagement but also echoes the Control-Value Theory (CVT) (Pekrun & Perry, 2014) that indicates the crucial effect of positive emotions in catalyzing learners' growth in L2 mindset, thus leading to language learning behaviors within traditional instructions or beyond (Liu et al., 2025a; 2025b; Wang et al., 2021; Zou et al., 2025a). On this ground, we find that the intricate impacts of the growth mindset and enjoyment on AI-IDLE are neither linear nor discrete, aligning with the insights of prior studies (Zhong et al., 2024) in terms of the complexity of enjoyment in connection with the growth mindset. In addition, it should be noted that although the result related to the impact of the growth L2 mindset on AI-IDLE did not reach conventional statistical significance ( $p = .054$ ), the trend suggests a potential effect that warrants further investigation with larger samples.

Furthermore, the study indicated that a stronger sense of L2 anxiety enabled greater investment in AI-IDLE. This observation diverges from the study made by Dewaele et al. (2023) that foreign language learners with a stronger sense of anxiety are less likely to improve their L2 achievement and engage in learning activities. One possible explanation is that the digital landscape of AI-IDLE may offer a less stressful and intimidating environment where learners can engage with language activities at their own pace without fear of immediate judgment or peer pressure, compared to traditional face-to-face classroom contexts (Lee & Drajati, 2019; Zou et al., 2025b). However, the study displayed that Pakistani multilingual youths' enjoyment made a positive and direct impact on AI-IDLE, which was supported by Hypothesis 6. This insight not only echoes Liu et al.'s (2025a; 2025b) findings that suggest Chinese EFL students' enjoyment serves as a powerful reinforcement for AI-IDLE practices but also aligns with Krashen's Affective Filter Hypothesis (Krashen, 1986) that implies positive emotions like enjoyment can lower the affective filter, allowing learners to be more receptive to language input and more willing to take risks in their language productions or behaviors, thereby facilitating more effective self-directed digital learning activities with the affordances of AI technologies. Within the CVT framework, learners who enjoy their learning are more likely to regard AI-IDLE as intrinsically rewarding, thereby enhancing their engagement with AI-IDLE activities. By contrast, the study demonstrated that L2 boredom lacks a predictive power over AI-IDLE,



which deviates from previous studies (Dai & Wang, 2024; Dewaele et al., 2023). One possible reason is that the ever-expanding range of AI-IDLE resources available may provide learners with a constant stream of new and diverse content, which could help maintain learners' curiosity and interest, thus reducing the impact of boredom on their AI-IDLE engagement (Wu et al., 2025).

### ***5.3 Pedagogical implications***

In line with this study, the central pedagogical implications for AI-mediated language teaching include several strategies that are somehow tied to all others. **First**, considering the direct influence of the growth L2 mindset on L2 enjoyment in AI-mediated language learning contexts, it would be valuable if teachers help students set achievable and realistic goals for their language learning and help them break down larger goals into smaller and manageable tasks to provide a sense of accomplishment. For instance, instead of setting a goal to “become fluent in English”, teachers encourage students to focus on specific objectives like “have a 5-minute conversation with an L1 English speaker.” **Second**, given the power of L2 enjoyment as a catalyst for AI-IDLE engagement, instructors may use AI-mediated technologies, such as AI chatbots or Virtual Reality (VR), to create an immersive and interactive language communication environment. For example, teachers set up a virtual exchange programme where students can communicate with English language learners from other countries using AI-mediated platforms. Cultivating authentic communication environments enables to increase students' enjoyment and foster their proactive engagement with language learning activities beyond the classroom (Zou et al., 2025a). **Third**, concerning the indirect and positive impact of the growth L2 mindset on AI-IDLE through the pathway of L2 enjoyment, practitioners and teachers may provide specific and constructive feedback that highlights the process of learning and strategies used, rather than solely focusing on outcomes. This approach can help foster a growth mindset and promote L2 enjoyment, thereby facilitating AI-IDLE engagement. For instance, rather than simply marking errors, teachers can provide specific guidance on how to improve and praise learners' efforts in engaging in language learning activities beyond the classroom. **Lastly**, an important prescription would be to create a scenario where students can freely express themselves, share in the successes, learn from the mistakes, and get rewarded for the effort (e.g., reward system), in turn enhancing enjoyment, lessening anxiety and boredom, and promoting active engagement outside of the classroom with AI-mediated learning tools.

## **6. Conclusion, limitations, and suggestions for future studies**

This study found that a growth L2 mindset motivates Pakistani multilingual youths to engage in AI-IDLE through enhanced L2 enjoyment, while negative emotions had no significant influence on this relationship. While these findings are valuable, this study is not free from limitations. First, this study used a cross-sectional research design to explore the relationships between the growth L2 mindset, emotions, and AI-IDLE, which fails to observe the longitudinal dynamics of these variables. Future research may consider employing a growth curve modeling approach to examine the longitudinal changes of these main variables in the long term. Second, we recommend future research can consider a larger sample size to reach more robust research findings. Third, given that the quantitative method employed in the present study has its limitations, we call for future research that can employ a mixed-method approach to achieve triangulation verification and capture nuanced patterns of these factors. Lastly, while our study contributed to Pakistani multilingual youths' growth L2 mindset,

emotions, and AI-IDLE, we urge scholars to explore the associations between motivational, emotional, and behavioural variables in other multilingual contexts.

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